

Hands-on Assignments for Using Single-Row Functions to Customize The Output

Complete the below hands-on assignments before proceeding with the next Topic

Update Completion Status

No.	Cat	Hands-on Assignment	Topics Covered
1	M	Write a query to display the current date. Label the column Date.	Date Functions

The screenshot displays the Oracle SQL Developer environment. The title bar reads "Oracle SQL Developer : Oracle". The interface includes a "Find Database Object" search bar, a tree view on the left showing the database structure with checkboxes for "All Schemas", "All Object Types", "All Identifier Types", "All Identifier Usages", "Columns", "Code", and "All Dependencies". The main workspace shows a worksheet with the SQL query: `SELECT SYSDATE AS "Date"` and `FROM DUAL;`. The execution time is displayed as "0.131 seconds". Below the worksheet, the "Script Output" pane shows the result: "Date" followed by a blank line and "29-06-26".

2	M	The HR department needs a report to display the employee number, last name, salary, and salary increased by 15.5% (expressed as a whole number) for each employee. Label the column New Salary.	Numeric Functions
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Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

☒ All Schemas
☒ All Object Types
☐ All Identifier Types
☐ All Identifier Usages
☐ Columns
☐ Code
☐ All Dependencies

Oracle hr_install.sql 0.138 seconds

Worksheet Query Builder

```

SELECT EMPLOYEE_ID,
       LAST_NAME,
       SALARY,
       ROUND(SALARY * 1.155, 0) AS "New Salary"
FROM HR.EMPLOYEES;
  
```

Script Output x

Task completed in 0.138 seconds

EMPLOYEE_ID	LAST_NAME	SALARY	New Salary
100	King	24000	27720
101	Yang	17000	19635
102	Garcia	17000	19635
103	James	9000	10395
104	Miller	6000	6930
105	Williams	4800	5544
106	Jackson	4800	5544
107	Nguyen	4200	4851
108	Gruenberg	12008	13869
109	Faviet	9000	10395
110	Chen	8200	9471
EMPLOYEE_ID	LAST_NAME	SALARY	New Salary
111	Sciarra	7700	8894
112	Urman	7800	9009
113	Popp	6900	7970
114	Li	11000	12705
115	Khoo	3100	3581
116	Baida	2900	3350
117	Tobias	2800	3234
118	Himuro	2600	3003
119	Colmenares	2500	2888
120	Weiss	8000	9240
121	Fripp	8200	9471

3

M

Modify the previous query to add a column alias that subtracts the old salary from the new salary. Label the column Increase

Column Alias

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

☒ All Schemas
☒ All Object Types
☐ All Identifier Types
☐ All Identifier Usages
☐ Columns
☐ Code
☐ All Dependencies

Worksheet Query Builder

```

SELECT EMPLOYEE_ID,
       LAST_NAME,
       SALARY,
       ROUND(SALARY * 1.155, 0) AS "New Salary",
       ROUND(SALARY * 1.155, 0) - SALARY AS "Increase"
FROM HR.EMPLOYEES;

```

Script Output x

Task completed in 0.125 seconds

EMPLOYEE_ID	LAST_NAME	SALARY	New Salary	Increase
100	King	24000	27720	3720
101	Yang	17000	19635	2635
102	Garcia	17000	19635	2635
103	James	9000	10395	1395
104	Miller	6000	6930	930
105	Williams	4800	5544	744
106	Jackson	4800	5544	744
107	Nguyen	4200	4851	651
108	Gruenberg	12008	13869	1861
109	Faviet	9000	10395	1395
110	Chen	8200	9471	1271
EMPLOYEE_ID	LAST_NAME	SALARY	New Salary	Increase
111	Sciarra	7700	8894	1194
112	Urman	7800	9009	1209
113	Popp	6900	7970	1070
114	Li	11000	12705	1705
115	Khoo	3100	3581	481
116	Baida	2900	3350	450
117	Tobias	2800	3234	434
118	Himuro	2600	3003	403
119	Colmenares	2500	2888	388
120	Weiss	8000	9240	1240
121	Fripp	8200	9471	1271

4

M

Write a query that displays the ename (with the first letter uppercase and all other letters lowercase) and the length of the ename for all employees whose name starts with the letters J, A, or M. Give each column an appropriate label. Sort the results by the employees' enames.

String Functions

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

☒ All Schemas
☒ All Object Types
☐ All Identifier Types
☐ All Identifier Usages
☐ Columns
☐ Code
☐ All Dependencies

Worksheet

Query Builder

```

SELECT INITCAP(LAST_NAME) AS "Name",
       LENGTH(LAST_NAME) AS "Length"
FROM HR.EMPLOYEES
WHERE UPPER(SUBSTR(LAST_NAME, 1, 1)) IN ('J', 'A', 'M')
ORDER BY LAST_NAME;
  
```

0.124 seconds

Script Output

Task completed in 0.124 seconds

Name	Length
Abel	4
Ande	4
Atkinson	8
Jackson	7
Jacobs	6
James	5
Johnson	7
Jones	5
Mallin	6
Markle	6
Marlow	6
Martinez	8
Marvins	7
Matos	5
Mcewen	6
McLeod	6
Mikkilineni	11
Miller	6
Mourgos	7

19 rows selected.

5

M

Rewrite the query so that the user is prompted to enter a letter that starts the last name. For example, if the user enters H when prompted for a letter, then the output should show all employees whose last name starts with the letter H.

Using Prompts and SQL Operators

Oracle SQL Developer : Oracle

Find Database Object

Oracle

☒ All Schemas
☒ All Object Types
☐ All Identifier Types
☐ All Identifier Usages
☐ Columns
☐ Code
☐ All Dependencies

Worksheet

hr_install.sql

4.67299986 seconds

```

SELECT LAST_NAME
FROM HR.EMPLOYEES
WHERE UPPER(SUBSTR(LAST_NAME, 1, 1)) = UPPER('&letter')
ORDER BY LAST_NAME;
  
```

Script Output

Task completed in 4.673 seconds

```

old:SELECT LAST_NAME
FROM HR.EMPLOYEES
WHERE UPPER(SUBSTR(LAST_NAME, 1, 1)) = UPPER('&letter')
ORDER BY LAST_NAME
new:SELECT LAST_NAME
FROM HR.EMPLOYEES
WHERE UPPER(SUBSTR(LAST_NAME, 1, 1)) = UPPER('2')
ORDER BY LAST_NAME
no rows selected
  
```

6

M

The HR department wants to find the length of employment for each employee. For each employee, display the ename and calculate the number of months between today and the date on which the employee was hired. Label the column MONTHS_WORKED. Order your results by the number of months employed. Round the number of months up to the closest whole number.

Date Functions

Oracle SQL Developer : Oracle

Find Database Object

Oracle

- ☒ All Schemas
- ☒ All Object Types
- ☐ All Identifier Types
- ☐ All Identifier Usages
- ☐ Columns
- ☐ Code
- ☐ All Dependencies

Worksheet | Query Builder

```
SELECT LAST_NAME,
       CEIL(MONTHS_BETWEEN(SYSDATE, HIRE_DATE)) AS MONTHS_WORKED
FROM   HR.EMPLOYEES
ORDER BY MONTHS_WORKED;
```

Script Output

Task completed in 0.1 seconds

LAST_NAME	MONTHS_WORKED
Kumar	99
Banda	99
Ande	100
Markle	100
Lee	101
Zlotkey	101
Geoni	101
Philtanker	101
Grant	102
Johnson	102
Marvins	102
LAST_NAME	MONTHS_WORKED
Perkins	103
Popp	103
Gee	103
Mourgos	104
Tuvault	104
Cambrault	105
Colmenares	107
Sullivan	109
OConnell	109
Grant	110
Miller	110

7

M

Create a report that produces the following for each employee: earns monthly but wants <3 times salary>. Label the column Dream Salaries.

Column Alias

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

☒ All Schemas
☒ All Object Types
☐ All Identifier Types
☐ All Identifier Usages
☐ Columns
☐ Code
☐ All Dependencies

Worksheet Query Builder

0.13600001 seconds

```

SELECT LAST_NAME || ' earns ' || SALARY || ' monthly but wants ' || (SALARY * 3) || '.' AS "Dream Salaries"
FROM HR.EMPLOYEES;
  
```

Script Output x

Task completed in 0.136 seconds

Dream Salaries

King earns 24000 monthly but wants 72000.
 Yang earns 17000 monthly but wants 51000.
 Garcia earns 17000 monthly but wants 51000.
 James earns 9000 monthly but wants 27000.
 Miller earns 6000 monthly but wants 18000.
 Williams earns 4800 monthly but wants 14400.
 Jackson earns 4800 monthly but wants 14400.
 Nguyen earns 4200 monthly but wants 12600.
 Gruenberg earns 12008 monthly but wants 36024.
 Favier earns 9000 monthly but wants 27000.
 Chen earns 8200 monthly but wants 24600.

Dream Salaries

Sciarra earns 7700 monthly but wants 23100.
 Urman earns 7800 monthly but wants 23400.
 Popp earns 6900 monthly but wants 20700.
 Li earns 11000 monthly but wants 33000.
 Khoo earns 3100 monthly but wants 9300.
 Baida earns 2900 monthly but wants 8700.
 Tobias earns 2800 monthly but wants 8400.
 Himuro earns 2600 monthly but wants 7800.
 Colmenares earns 2500 monthly but wants 7500.
 Weiss earns 8000 monthly but wants 24000.
 Fripp earns 8200 monthly but wants 24600.

8

M

Create a query to display the last name and salary for all employees. Format the salary to be 15 characters long, left-padded with the \$ symbol. Label the column SALARY.

String Functions

Oracle SQL Developer : Oracle

Find Database Object

Oracle

- ☒ All Schemas
- ☒ All Object Types
- ☐ All Identifier Types
- ☐ All Identifier Usages
- ☐ Columns
- ☐ Code
- ☐ All Dependencies

Worksheet Query Builder

```
SELECT LAST_NAME,  
       LPAD(TO_CHAR(SALARY), 15, '$') AS SALARY  
FROM HR.EMPLOYEES;
```

Script Output

Task completed in 0.159 seconds

LAST_NAME	SALARY
King	\$\$\$\$\$\$\$\$\$24000
Yang	\$\$\$\$\$\$\$\$\$17000
Garcia	\$\$\$\$\$\$\$\$\$17000
James	\$\$\$\$\$\$\$\$\$9000
Miller	\$\$\$\$\$\$\$\$\$6000
Williams	\$\$\$\$\$\$\$\$\$4800
Jackson	\$\$\$\$\$\$\$\$\$4800
Nguyen	\$\$\$\$\$\$\$\$\$4200
Gruenberg	\$\$\$\$\$\$\$\$\$12008
Faviet	\$\$\$\$\$\$\$\$\$9000
Chen	\$\$\$\$\$\$\$\$\$8200

LAST_NAME	SALARY
Sciarra	\$\$\$\$\$\$\$\$\$7700
Urman	\$\$\$\$\$\$\$\$\$7800
Popp	\$\$\$\$\$\$\$\$\$6900
Li	\$\$\$\$\$\$\$\$\$11000
Khoo	\$\$\$\$\$\$\$\$\$3100
Baida	\$\$\$\$\$\$\$\$\$2900
Tobias	\$\$\$\$\$\$\$\$\$2800
Himuro	\$\$\$\$\$\$\$\$\$2600
Colmenares	\$\$\$\$\$\$\$\$\$2500
Weiss	\$\$\$\$\$\$\$\$\$8000
Fripp	\$\$\$\$\$\$\$\$\$8200

9

M

Display each employee's last name, hire date, and salary review date, which is the first Monday after six months of service. Label the column REVIEW. Format the dates to appear in the format similar to "Monday, the Thirty-First of July, 2000."

Date Functions

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

All Schemas
All Object Types
All Identifier Types
All Identifier Usages
Columns
Code
All Dependencies

Worksheet Query Builder

```
SELECT LAST_NAME,
       HIRE_DATE,
       TO_CHAR(NEXT_DAY(ADD_MONTHS(HIRE_DATE, 6), 'MONDAY'),
              'Day, "the" Ddspth "of" Month, YYYY') AS REVIEW
FROM HR.EMPLOYEES;
```

0.198 seconds

Script Output

Task completed in 0.198 seconds

LAST_NAME	HIRE_DAT	REVIEW
King	17-06-13 Monday	, the Twenty-Third of December , 2013
Yang	21-09-15 Monday	, the Twenty-Eighth of March , 2016
Garcia	13-01-11 Monday	, the Eighteenth of July , 2011
James	03-01-16 Monday	, the Fourth of July , 2016
Miller	21-05-17 Monday	, the Twenty-Seventh of November , 2017
Williams	25-06-15 Monday	, the Twenty-Eighth of December , 2015
Jackson	05-02-16 Monday	, the Eighth of August , 2016
Nguyen	07-02-17 Monday	, the Fourteenth of August , 2017
Gruenberg	17-08-12 Monday	, the Eighteenth of February , 2013
Faviet	16-08-12 Monday	, the Eighteenth of February , 2013
Chen	28-09-15 Monday	, the Fourth of April , 2016
LAST_NAME	HIRE_DAT	REVIEW
Sciarra	30-09-15 Monday	, the Fourth of April , 2016
Urman	07-03-16 Monday	, the Twelfth of September, 2016
Popp	07-12-17 Monday	, the Eleventh of June , 2018
Li	07-12-12 Monday	, the Tenth of June , 2013
Khoo	18-05-13 Monday	, the Twenty-Fifth of November , 2013
Baida	24-12-15 Monday	, the Twenty-Seventh of June , 2016
Tobias	24-07-15 Monday	, the Twenty-Fifth of January , 2016
Himuro	15-11-16 Monday	, the Twenty-Second of May , 2017
Colmenares	10-08-17 Monday	, the Twelfth of February , 2018
Weiss	18-07-14 Monday	, the Nineteenth of January , 2015
Frapp	10-04-15 Monday	, the Twelfth of October , 2015
LAST_NAME	HIRE_DAT	REVIEW

10

M

Display the last name, hire date, and day of the week on which the employee started. Label the column DAY. Order the results by the day of the week, starting with Monday.

Date Functions

Find Database Object

Oracle

Go

☒ All Schemas
☒ All Object Types
☐ All Identifier Types
☐ All Identifier Usages
☐ Columns
☐ Code
☐ All Dependencies

Oracle hr_install.sql

0.16599999 seconds

Worksheet Query Builder

```

SELECT LAST_NAME,
       HIRE_DATE,
       TO_CHAR(HIRE_DATE, 'Day') AS DAY
FROM HR.EMPLOYEES
ORDER BY CASE TO_CHAR(HIRE_DATE, 'FMDAY', 'NLS_DATE_LANGUAGE=ENGLISH')
          WHEN 'MONDAY' THEN 1
          WHEN 'TUESDAY' THEN 2
          WHEN 'WEDNESDAY' THEN 3
          WHEN 'THURSDAY' THEN 4
          WHEN 'FRIDAY' THEN 5
          WHEN 'SATURDAY' THEN 6
          WHEN 'SUNDAY' THEN 7
END;

```

Script Output

Task completed in 0.166 seconds

LAST_NAME	HIRE_DAT	DAY
King	17-06-13	Monday
Davis	17-08-15	Monday
Yang	21-09-15	Monday
Martinez	17-02-14	Monday
Feeney	23-05-16	Monday
Venzl	11-07-16	Monday
Sarchand	27-01-14	Monday
Zlotkey	29-01-18	Monday
Partners	05-01-15	Monday
Stiles	26-10-15	Monday
Olson	10-04-17	Monday
LAST_NAME	HIRE_DAT	DAY
Marlow	16-02-15	Monday
Urman	07-03-16	Monday
Chen	28-09-15	Monday
Bell	04-02-14	Tuesday
Nguyen	07-02-17	Tuesday
Sullivan	04-03-14	Tuesday

11 M

Create a query that displays the employees' last names and commission amounts. If an employee does not earn commission, show "No Commission." Label the column COMM.

Other Functions

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

All Schemas
All Object Types
All Identifier Types
All Identifier Usages
Columns
Code
All Dependencies

Worksheet Query Builder

0.168 seconds

```
SELECT LAST_NAME,
       NVL(TO_CHAR(COMMISSION_PCT), 'No Commission') AS COMM
FROM HR.EMPLOYEES;
```

Script Output x

Task completed in 0.168 seconds

LAST_NAME	COMM
King	No Commission
Yang	No Commission
Garcia	No Commission
James	No Commission
Miller	No Commission
Williams	No Commission
Jackson	No Commission
Nguyen	No Commission
Gruenberg	No Commission
Faviet	No Commission
Chen	No Commission

LAST_NAME	COMM
Sciarra	No Commission
Urman	No Commission
Popp	No Commission
Li	No Commission
Khoo	No Commission
Baida	No Commission

12

M

Create a query that displays the first eight characters of the employees' last names and indicates the amounts of their salaries with asterisks. Each asterisk signifies a thousand dollars. Sort the data in descending order of salary. Label the column EMPLOYEES_AND_THEIR_SALARIES.

String Functions

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

Worksheet

Query Builder

```
SELECT SUBSTR(LAST_NAME, 1, 8) || LPAD('*', ROUND(SALARY / 1000), '*') AS "EMPLOYEES_AND_THEIR_SALARIES"
FROM HR.EMPLOYEES
ORDER BY SALARY DESC;
```

Script Output

Task completed in 0.147 seconds

EMPLOYEES_AND_THEIR_SALARIES
King*****
Yang*****
Garcia*****
Singh*****
Partners*****
Martinez*****
Gruenber*****
Higgins*****
Errazuri*****
Ozer*****
Li*****
EMPLOYEES_AND_THEIR_SALARIES
Cambrault*****
Abel*****
Zlotkey*****
Vishney*****
King*****
Tucker*****
Brown*****
Bloom*****
Fox*****
Greene*****
Bernstei*****
EMPLOYEES_AND_THEIR_SALARIES

13

M

Using the DECODE function, write a query that displays the grade of all employees based on the value of the column JOB_ID, using the following data: PRESIDENT-A, MANAGER-B, SALESMAN-C, CLERK-D

Other Functions

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

Oracle hr_install.sql 0.222 seconds

Worksheet Query Builder

```

SELECT LAST_NAME,
       JOB_ID,
       DECODE(JOB_ID,
              'AD_PRES', 'A',
              'AD_VP', 'B',
              'SA_MAN', 'B',
              'SA_REP', 'C',
              'ST_CLERK', 'D',
              'SH_CLERK', 'D',
              'HR_CLERK', 'D',
              'PU_CLERK', 'D',
              'IT_PROG', 'D',
              'Unknown') AS GRADE
FROM HR.EMPLOYEES;

```

Script Output x

Task completed in 0.222 seconds

LAST_NAME	JOB_ID	GRADE
Abel	SA_REP	C
Ande	SA_REP	C
Atkinson	ST_CLERK	D
Baida	PU_CLERK	D
Banda	SA_REP	C
Bates	SA_REP	C
Bell	SH_CLERK	D
Bernstein	SA_REP	C
Bissot	ST_CLERK	D
Bloom	SA_REP	C
Brown	PR_REP	Unknown

LAST_NAME	JOB_ID	GRADE
Bull	SH_CLERK	D
Cabrio	SH_CLERK	D
Cambrault	SA_MAN	B

Oracle SQL Developer : Oracle

Find Database Object

Oracle

☒ All Schemas
☒ All Object Types
☐ All Identifier Types
☐ All Identifier Usages
☐ Columns
☐ Code
☐ All Dependencies

Go

Oracle hr_install.sql 0.14 seconds

Worksheet Query Builder

```
SELECT LAST_NAME,  
       JOB_ID,  
       CASE JOB_ID  
         WHEN 'AD_PRES' THEN 'A'  
         WHEN 'AD_VP' THEN 'B'  
         WHEN 'SA_MAN' THEN 'B'  
         WHEN 'SA_REP' THEN 'C'  
         WHEN 'ST_CLERK' THEN 'D'  
         WHEN 'SH_CLERK' THEN 'D'  
         WHEN 'HR_CLERK' THEN 'D'  
         WHEN 'PU_CLERK' THEN 'D'  
         WHEN 'IT_PROG' THEN 'D'  
         ELSE 'Unknown'  
       END AS GRADE  
FROM HR.EMPLOYEES;
```

Script Output

Task completed in 0.14 seconds

LAST_NAME	JOB_ID	GRADE
Abel	SA_REP	C
Ande	SA_REP	C
Atkinson	ST_CLERK	D
Baida	PU_CLERK	D
Banda	SA_REP	C
Bates	SA_REP	C
Bell	SH_CLERK	D
Bernstein	SA_REP	C
Bissot	ST_CLERK	D
Bloom	SA_REP	C
Brown	PR_REP	Unknown
LAST_NAME	JOB_ID	GRADE
Bull	SH_CLERK	D
Cabrio	SH_CLERK	D
Cambrault	SA_MAN	B